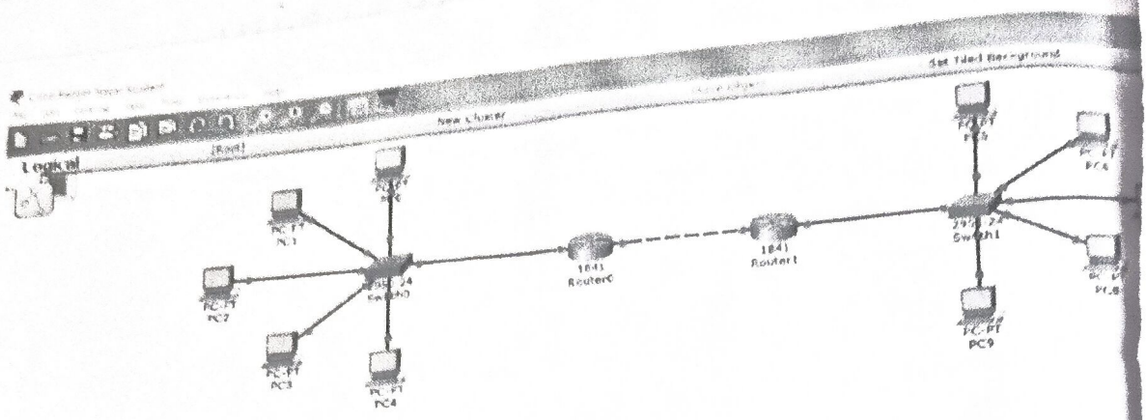
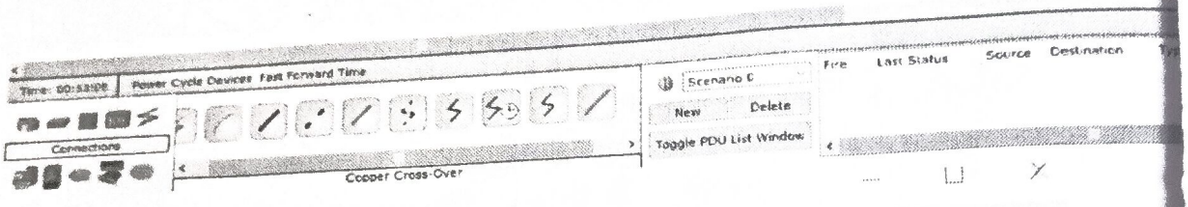




Router0

Physical | Config | CLI

IOS Command Line Interface

```
exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 192.168.1.1 255.255.255.224
% 192.168.1.0 overlaps with FastEthernet0/0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.65 255.255.255.224
% 192.168.1.64 overlaps with FastEthernet0/1
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#do show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol

FastEthernet0/0        192.168.1.1     YES manual up                    down

FastEthernet0/1        192.168.1.65    YES manual up                    up

Vlan1                  unassigned      YES unset  administratively down down
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to
up
```

Copy | Paste

11/9/26

Exp:09 Implementation of Subnetting IN CISCO Packet tracer Simulator.

Aim:

To Implement Classless IP subnetting using cisco packet tracer and configure the network devices with appropriate IP addresses and Subnet Masks to establish connection b/w different Subnets.

Procedure:

(1) In the CiscoPacket tracer create a new network topology

(2) Add the required network devices.

⇒ 2 routers (R_0 and R_1)

⇒ 2 Switches (S_0 and S_1)

⇒ To Pc's (PC_0 - PC_9)

(3) Connect PC_0 - PC_4 to switch S_0 and PC_5 - PC_9 to switch S_1

(4) connect S_0 to R_0 and S_1 to R_2

(5) connect R_0 and R_1 using their availabity. Ethernet.

(6) Divide the Network into 1027 Subnets. the Submit Mask used is 255.255.255.224

(7) Enable the router Interface using no Shutdown Simulator Command

(8) Enable the R_1 Infaces using the no shutdown Command to R_0

(9) Configure the IP addresses of PC_0 - PC_4 in 1st Subnet

a) PC_0 - 192.168.1.11 (b) PC_1 - 192.168.1.12

Command Prompt

```
PC>ping 192.168.1.33

Pinging 192.168.1.33 with 32 bytes of data:

Reply from 192.168.1.33: bytes=32 time=83ms TTL=255
Reply from 192.168.1.33: bytes=32 time=0ms TTL=255
Reply from 192.168.1.33: bytes=32 time=0ms TTL=255
Reply from 192.168.1.33: bytes=32 time=0ms TTL=255

Ping statistics for 192.168.1.33:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 83ms, Average = 20ms

PC>ping 192.168.1.42

Pinging 192.168.1.42 with 32 bytes of data:

Reply from 192.168.1.42: bytes=32 time=1ms TTL=128
Reply from 192.168.1.42: bytes=32 time=0ms TTL=128
Reply from 192.168.1.42: bytes=32 time=0ms TTL=128
Reply from 192.168.1.42: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.42:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

PC>
```

Command Prompt

```
PC>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=16ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=11ms TTL=128
Reply from 192.168.1.11: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 16ms, Average = 8ms


PC>ping 192.168.1.12

Pinging 192.168.1.12 with 32 bytes of data:

Reply from 192.168.1.12: bytes=32 time=41ms TTL=128
Reply from 192.168.1.12: bytes=32 time=1ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128
Reply from 192.168.1.12: bytes=32 time=0ms TTL=128

Ping statistics for 192.168.1.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 41ms, Average = 10ms

PC>
```

c) PC_2 - 192.168.1.13

d) PC_3 - 192.168.1.14

e) PC_4 - 192.168.1.15

(10) Configure static routes on R_0 and R_1 so that packets can travel b/w the LAN subnets

(11) Verify the router interfaces using the command "show IP interface brief"

(12) Test connectivity b/w PCs using the Ping command from CMD.

(13) Ping the default gateway from each PC and then Ping PC's belonging to other subnet

(14) If successful replies are recieved with 0% packet loss, the subnetted network is functioning correctly.

Result:

Thus, classless IP subnetting was successfully implemented using Packet tracer and connectivity between the subnetted networks was verified using Ping command.